

Smartphone addiction among children, adolescents and teenagers: mapping emerging and future direction

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ABSTRACT

This study explores the fundamental knowledge structure of smartphone addiction among young consumers. Smartphone use has been ubiquitous and a norm in today's society. This phenomenon led to smartphone addiction among users, particularly youngsters, as it imparts negative consequences to their growth, development, and social life. Employing a novel state-of-the-art science mapping approach through bibliometric analysis, the current and future trends of smartphone addiction among young consumers were analyzed. 665 documents were retrieved from the Web of Science (WoS) database and were analyzed using bibliographic coupling and co-word analysis. Four emerging research streams produced: 1) Negative consequences of smartphone addiction among youngsters, 2) Parental phubbing and its impact on adolescent smartphone addiction, 3) Smartphone and social media addiction, and 4) Smartphone addiction scale. At the same time, future trends are related to the risk and impact of smartphone addiction among young consumers.

1. Introduction

Smartphones have benefited society through the ease of communication, entertainment and all-in-one applications. With these multi-function abilities replacing various other gadgets, it has become a critical device in human life [1]. The number of smartphone users has increased steadily in the past decades. Today, there are approximately 7.21 billion smartphones, accounting for 90 % of the 8 billion global population [2]. As smartphone coverage increases, children are easily exposed in the early months of life [3]. For the young generation in their adolescence and teenagerhood, smartphones provide the means to connect with friends, interact, communicate and as a means for entertainment. Excessive smartphone use negatively impacts their relationship with their parents and children, their studies, emotional development, and personal safety [4].

People are becoming more attached to smartphones in such a way that it disrupts their daily lives and makes them anxious when separated. This habitual use of smartphones leads to smartphone addiction, or it can be described in other terminology, such as mobile phone addiction or problematic smartphone use [5]. Children suffer the most from this habitual use of smartphones. According to well-accepted adoption theories, such as the technology acceptance model (TAM) and the unified

theory of acceptance and use of technology (UTAUT), smartphone adoption among youngsters is mainly due to their perception of usefulness, ease of use, and the influence of peers and social networks, supported by facilitating conditions. These factors align with young users' lifestyles as practical communication and learning tools and essential devices for social interaction and entertainment [6]. Furthermore, children's growth, development, and wellbeing are impacted when parents use smartphones and neglect their attention. This is phubbing (phone + snubbing) [7]. Parental phubbing is the cause of smartphone addiction in adolescents [8,9]. Phubbing occurs because of smartphone, social media, game, or internet addiction. This social problem is a matter of concern in today's society, and careful intervention and mitigation strategies are needed.

This study is motivated by two reasons. First, smartphone addiction is a serious problem, impacting families and societies. It is a big issue impacting the core of the family level. Children are exposed to smartphones as early as months after birth, disrupting their mental and physical development [3]. Next, the adolescence stage is probably the most fragile age group (between 10 and 19 years old). It is a transitional period from childhood to adulthood, comprising substantial development and growth [10]. Children entering the adolescence stage experience stronger peer identity and the new freedom that may lead to risky

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behavior of smartphone use [11]. The rapid development of smartphone technology has led to easier adoption due to their perceived ease of use and security factors [12]. These factors collectively highlight the urgent need to understand the drivers of smartphone addiction among adolescents in order to develop effective preventive and intervention strategies. Teenagers, despite being of a more mature age and having particular concerns about smartphone overuse, are often optimistic regarding their online activities, underestimating the negative impacts of smartphone usage [13]. The differences in these age groups related to smartphone addiction warrant a detailed exploration of the knowledge structure based towards understanding the basis and fundamental insight. The findings would benefit relevant stakeholders, including policymakers, practitioners, scholars and parents, as the closest entity to these minors.

Second, concerning the review approach, few studies have looked into the specific scope of smartphone addiction through science mapping analysis. Despite that, several studies have provided a fundamental landscape of the subject. For instance, Fischer-Grote et al. [14] discovered that most studies on problematic smartphone use in adolescents negatively correlate with wellbeing, life satisfaction, and quality of life. In another study, Olson et al. [15], in their meta-analysis, discovered that problematic smartphone use is increasing worldwide. A sample collected from 24 countries from 33,831 respondents shows that smartphone addiction will continue to increase in the coming years. Khan and Khan [1] reviewed the global smartphone addiction studies, and one of the themes discovered was smartphone addiction among children and teenagers. M'endez et al. (2024) reviewed smartphone addiction in adolescent cognition based on functional magnetic resonance imaging studies. The findings show that there are neuroanatomical similarities between smartphone addiction in adolescents associated with reward processing and executive functions. Ge et al. [16] performed a systematic review and meta-analysis of studies on the relationship between loneliness and smartphone/internet addiction. There was a moderate positive association between loneliness and smartphone or internet addiction for 21 reviewed studies. In another study, Tok and Cumurcu [10] studied adolescents' addiction and discovered that smartphone and internet addiction are among the most prevalent addictions.

These past studies serve as the fundamental overview of smartphone addiction. However, there is a pertinent gap in addressing the aggregation of this smartphone addiction issue among these young consumers comprising children, adolescents and teenagers. Furthermore, understanding Generation Z's digital behaviors is crucial, as their adoption of digital products is shaped by factors such as social influence, digital self-efficacy, perceived cost, and compatibility [17]. Addressing this gap can provide deeper insights into how these young consumers are vulnerable to developing unhealthy smartphone habits. As such, this study contributes to the body of knowledge by presenting a science mapping of the subject through exploring the temporal and topological structure using bibliometric analysis. This study presents the following objectives:

1. To analyze the current and emerging trends of smartphone addiction among young consumers using bibliographic coupling
2. To explore future trends of smartphone addiction among young consumers using co-word analysis

This review is constructed as follows. This section presents an overview of smartphone addiction, motivation and objectives. Section 2 outlines the methodology of bibliometric analysis through a science mapping approach. Next, section 3 presents the findings and discussion based on the two science mapping analyses. Section 4 presents the theoretical and managerial implications. Section 5 discusses the limitations and future research avenues. Finally, section 6 concludes the study.

2. Methodology

2.1. Bibliometric approach

The bibliometric approach is a novel state-of-the-art quantitative method that analyzes large bibliographic databases using a science mapping technique Donthu et al. [18]. The approach has been accepted as complementary to the systematic review approach and meta-analysis [19]. Bibliometric analysis can be categorized into performance analysis and science mapping [20]. The former analyzes the contributions of research constituents based on authors, journals (sources), institutions and countries Donthu et al. [18]. The most prominent measures are the number of publications as the proxy of productivity and citations as the indicator of impact and influence. In the latter, science mapping synthesizes past findings as the basis to advance the line of research by mapping the structure and development of a particular field [21]. To extend the fundamental knowledge structure of the subject, this study employs science mapping analysis. Two popular science mapping approaches are utilized: 1) bibliographic coupling to analyze current and emerging themes and 2) co-word analysis to forecast future research streams of a field.

2.2. Research design and data collection procedure

The following search string (Table 1) was applied to retrieve publications based on relevant associated keywords deemed to be closely related. The keywords are related to smartphone addiction and young consumers comprising associated terminology such as adolescent, children, teenager and minor. The Web of Science (WoS) topic search (TS) function was applied to retrieve publications from the WoS core collection. Only journal publications were included in the analysis, leaving books, book chapters, conference proceedings, and editorials. Including journal publications ensures that only quality peer-reviewed publications are included in the analysis. Many studies have applied only to journal publications for the same reason [22,23,24]. Furthermore, to capture recent streams, the time horizon was limited to the last five years, from 2019 to 2024. To visualize the network map, VOSviewer version 1.6.18 was applied. The software has great visualization capability, uploading and exporting information from various sources [25].

3. Findings and discussion

The WoS core collection database was searched on 28th November 2024. The initial retrieved documents were 734. After limiting to only journal publications within the last six years, 665 was finalized and proceeded to descriptive analysis. Since the subject is dynamic with many recent developments, it is crucial to capture the most recent streams to ensure that the analysis reflects the latest trends and provides an up-to-date understanding of the research landscape. The number of citations was 11,584 and 9,753 (without self-citation). The number of publications and citations are presented in Fig. 1. Overall, the trend shows that it is increasing, peaking in 2022. Research on smartphone addiction was the highest in 2022, contributing to smartphone addiction studies during the COVID-19 pandemic. Despite declining the following year, it is expected to increase in the coming years as this is a global issue

Table 1
Search string in WoS database.

No	Keywords	Justification
1	"smartphone addiction*" OR "mobile* addiction"	To identify literature related to smartphone and mobile phone addiction
2	"adolescent*" OR "young*" OR "youth*" OR "child*" OR "minor*" OR "teen"	To identify literature related to young consumers

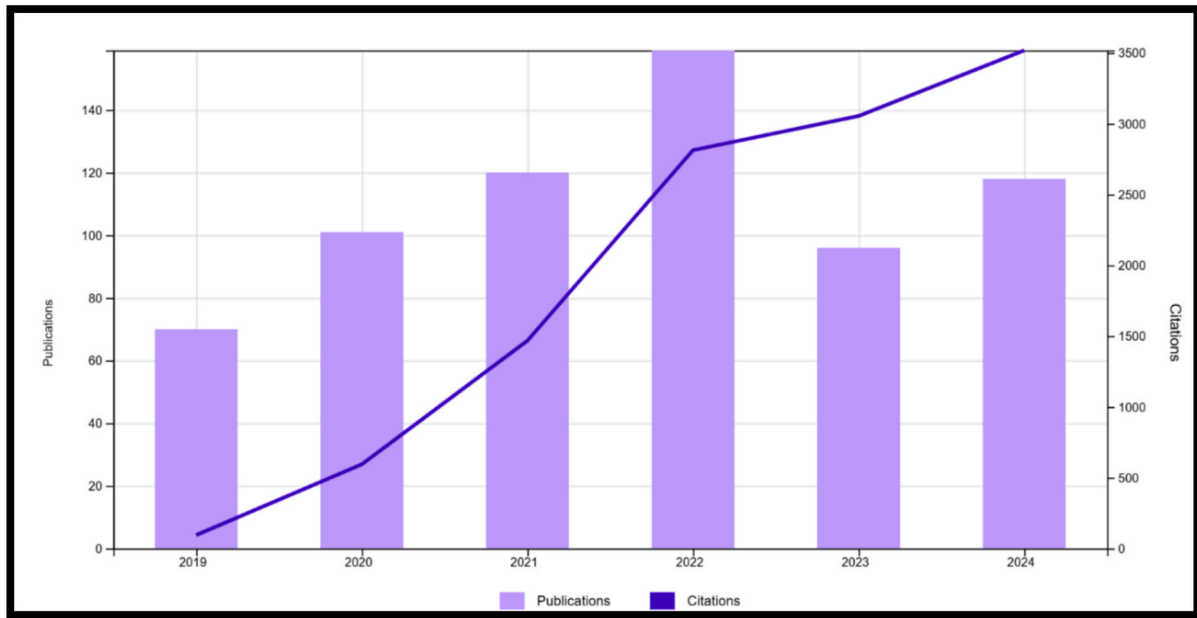


Fig. 1. Number of publications and citations on smartphone addiction among youngsters ().

Source: Web of Science

impacting society at large.

3.1. Bibliographic coupling analysis

Of the 665 documents, 52 met a threshold of 52 minimum citations. After several trials, the citation threshold was determined to determine the most suitable for producing a robust network map. The map must be appropriate with the appropriate citation threshold. It should not be too high, causing overfiltering and leading to missing out on important clusters in the network. Conversely, setting it too low may cause underfiltering, resulting in redundant clusters in the network map [26,27]. Since bibliographic coupling measures the relationship within the citing publication, the total link strength (TLS) is the interest value. Hence, the top 10 documents are shown in Table 2.

Fig. 2 illustrates the network visualization of the bibliographic coupling analysis. The four clusters appear distinctly separate, indicating four thematic groupings. The authors analyze these themes through inductive and qualitative interpretation.

- Cluster 1 (red): **Negative consequences of smartphone addiction among youngsters**

It is well-accepted that smartphone addiction possesses many negative impacts on young consumers, as reflected in this cluster. Zou et al. [38] discovered that smartphone addiction poses the risk of high blood pressure among adolescents. The prevalence is also associated with poor sleep quality and obesity. Kartal and Yabancı Ayhan [39] found that within a sample of college students, female was found to have a higher risk of eating disorders compared to males. The prevalence of smartphone addiction and internet addiction is higher among students with eating disorders. Mustafaoglu et al. [40] discovered that smartphone addiction is associated with a musculoskeletal disorder. The body parts reported with the highest musculoskeletal pain were the upper back, neck and wrists/hands. It is prevalent that the most emerging themes of recent publications suggest that smartphone addiction causes several negative implications for young consumers.

- Cluster 2 (green): **Parental phubbing and impact on adolescent smartphone addiction**

Table 2

Top 10 documents based on total link strength.

Rank	Publication		Citation	TLS
	Durak [28].	Predictors of nomophobia and smartphone addiction among adolescents in Turkey	111	302
	Mahapatra [29].	Loneliness and self-regulation as consequences of smartphone addiction	139	271
	Barnes et al. [30].	Relationship smartphone addiction and social network services.	113	236
	Yang et al. [31].	Smartphone use and its relation to academic procrastination, subjective wellbeing, academic anxiety and self-regulation	123	235
	Buctot et al [32].	The predictive capacity of smartphone addiction prevalence among Filipino adolescents.	58	214
	Li et al. [33].	The relation between fear of missing out (FoMO) and smartphone addiction.	74	200
	Ivanova et al. [34].	Depression among men and women and relationship with mobile phone addiction and phubbing.	72	189
	Li et al. [35].	FoMO and smartphone addiction among Chinese university students.	61	175
	Liu et al. [36].	Mobile phone addiction in adolescent: Peer victimization, self-compassion and gender	63	175
	Jeong et al. [37].	The difference between smartphone addiction and Internet addiction	62	165

Parents are the children's first role models. Their susceptible attachment to smartphones depends on how parents treat them at home and the impact of their relationship. Niu et al. [41] discovered that the parent-child relationship mediated the relationship between adolescent problematic smartphone use and parental phubbing. Hong et al. [49] show that parents phubbing not only impacted children's problematic smartphone use but also their self-esteem. This cluster suggests that there must be a mechanism to control parental phubbing and strategic intervention to reduce problematic smartphone use among adolescents. Xie et al. [42] demonstrated that parents' phubbing was positively

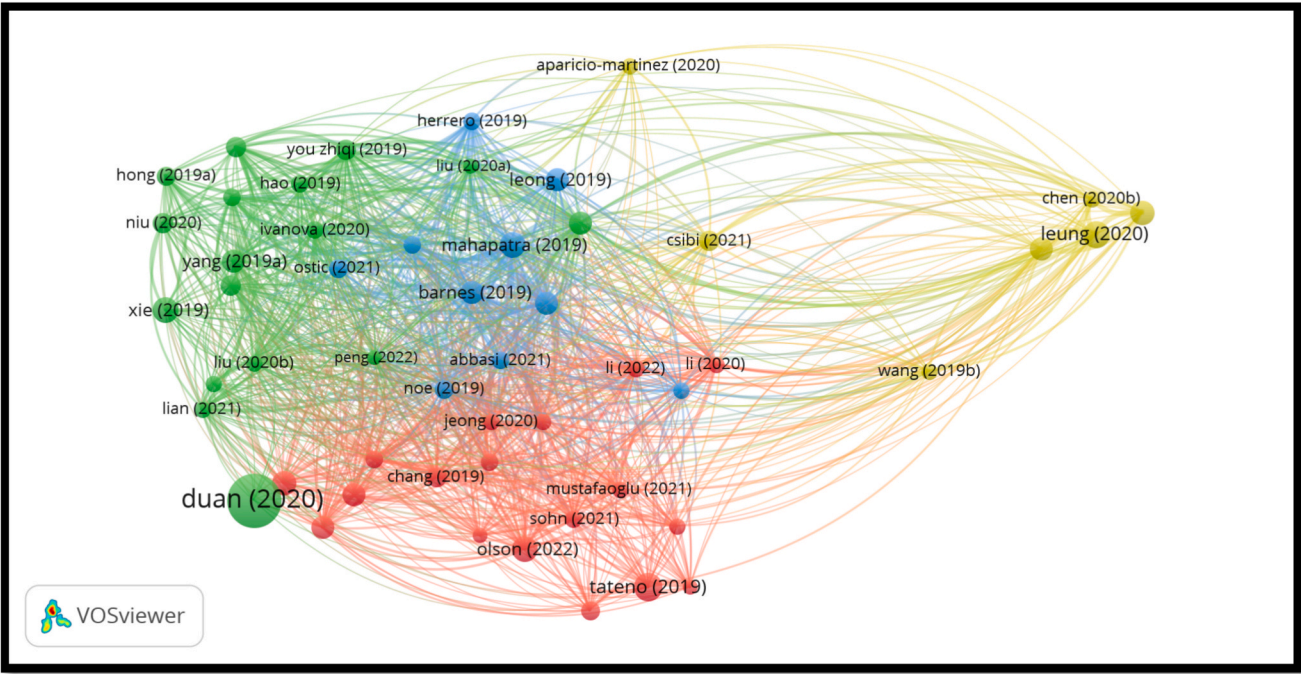


Fig. 2. Bibliographic coupling of smartphone addiction among youngsters.

associated with adolescents’ smartphone addiction. This relationship is further mediated by parent–child attachment and deviant peers, indicating that prior engagement, internal and external, bears an impact on adolescent smartphone addiction. As a crucial predictor of children’s smartphone addiction, parents’ use of smartphones or parental phubbing is an area of concern in understanding children’s smartphone addiction.

• Cluster 3 (blue): **Smartphone and social media addiction**

Addiction to social media and smartphones in today’s digital world goes together. One usually is addicted to social media applications through smartphones as the medium. Entertainment and game-related activity on social networking sites has led to youngsters’ smartphone addiction [43]. Barnes et al. [30] examined the relationship between smartphone and social media addiction among business students in the United States. Findings suggest that smartphone addiction is greater than social networking addiction, varied by educational attainment. Leong et al. [44] studied social media addiction and its impact on students’ Big Five personalities. It was discovered that apart from entertainment, neuroticism and agreeableness were the strongest predictors of social media addiction. Ostic et al. [45] point out the impact of social media towards smartphone addiction and users’ psychological well-being. Among youngsters, excessive use of social media was associated with fear of missing out.

• Cluster 4 (yellow): **Smartphone addiction scale**

Since smartphone addiction has become a crucial phenomenon towards society, researchers have come up with various smartphone addiction scales to measure behavior correctly. Yam et al. [46] applied three scales: the Bergen Social Media Addiction Scale (BSMAS), the Smartphone Application-Based Addiction Scale (SABAS) and the Internet Gaming Disorder Scales-Short Form (IGDS-SF9) on Hong Kong University students. All three scales were discovered to be unidimensional with satisfactory indices. Leung et al. [47] studied Taiwan and Hong Kong university students using the same three scales. The three scales were found to assess internet-related addiction among higher

education students validly. In the primary school context, Chen et al. [48] discovered that the three scales produced a unidimensional structure and were invariant across genders. These various smartphone addiction scales indicate the crucial prevalence of the smartphone addiction phenomenon in today’s society.

Table 3 summarizes the bibliographic coupling analysis, including cluster number and colour, labels, number of publications, and representative publications.

3.2. Co-word analysis

The co-word analysis predicts the future trends in smartphone addiction and young consumers in this study. A robust network map was finalized at the 28 occurrences threshold, with 48 out of 2,119 keywords included. The highest co-occurred keywords are smartphone addiction (460 occurrences), adolescents (318 occurrences) and depression (229 occurrences) (see Table 4). The main keyword shows that smartphone addiction is associated with adolescents and its impact, such as depression, anxiety and stress. Furthermore, apart from smartphone addiction, other related maladaptive usages are internet and mobile phone addiction.

Table 3
Summary of recent network clusters on smartphone addiction among young consumers.

Cluster No and colour	Cluster label	Number of publications	Representative publication
1 (red)	Negative consequences of smartphone addiction among youngsters	18	Zou et al. [38], Kartal and Ayhan [39] [40]
2 (green)	Parental phubbing and impact on adolescent smartphone addiction	17	[41], Hong et al. [49], Xie et al. [42]
3 (blue)	Smartphone and social media addiction	10	[43,30,44,45]
4 (yellow)	Smartphone addiction scale	7	[48,47,46]

Table 4
Top 15 keywords in smartphone addiction and young consumers.

Rank	Keyword	Occurrences	TLS
	Smartphone addiction	460	2215
	Adolescents	318	1629
	Depression	229	1331
	Anxiety	217	1249
	Internet addiction	186	985
	Addiction	113	528
	Stress	112	668
	Mobile phone addiction	110	639
	Prevalence	80	417
	Problematic smartphone use	79	410
	Symptoms	76	466
	Mobile phone	76	396
	Sleep quality	68	401
	Students	65	333
	Internet	62	329

Fig. 3 depicts the network structure of the co-word analysis, clearly displaying four clusters that represent distinct themes. Based on the author’s inductive interpretation, appropriate labels are assigned to each cluster.

• Cluster 1 (red): **The Impact of Smartphone Addiction on Adolescents**

Smartphone addiction impacts adolescents’ functioning. The adolescent stage requires adequate parental supervision as they are being exposed to gadgets in their daily lives. They will experience troubled psychological adjustment to the world outside their screen. In their study, Mohta and Halder [50] discovered that adolescents exposed to smartphones scored low in impulsivity, working memory and social

desirability. Fischer-Grote et al. [14], in their systematic review of 9 studies, discovered that, except for one, problematic smartphone use leads to negative life satisfaction, quality of life and wellbeing. Another impact of smartphones on adolescents is loneliness [16]. Adolescents who are attached to smartphones will lose social interaction outside due to prolonged attachment to smartphones. This phenomenon is a serious concern for developing mitigation and intervention strategies at the family level.

• Cluster 2 (green): **Risk factors and symptoms of smartphone addiction in children**

Children are one of the most vulnerable groups in problematic smartphone use. They are much younger than adolescents, ranging from 2-12 years old. Excessive use of smartphones among children leads to poor sleep quality, poor vision and psychological dependence [51]. Yadav and Chakraborty [3] discovered that smartphone overuse led to obesity and sleeping disorders among children. Mokhtarinia et al. [52] present the impact of smartphone addiction on children’s musculo-skeletal discomfort. The prevalence of smartphone addiction leads to discomfort in the eyes, neck, shoulders, upper back and wrists. These findings suggest that smartphone addiction in children is caused by caregivers’ or parents’ inability to control their smartphone usage. Implementing stern regulations on using handheld devices would mitigate smartphone addiction from its root.

• Cluster 3 (blue): **Smartphone Addiction and Higher Education Students: Depression, Anxiety, and Self-Esteem**

Smartphone addiction is highly prevalent among higher education students. Yilmaz et al. [53] revealed that nomophobia had a significant

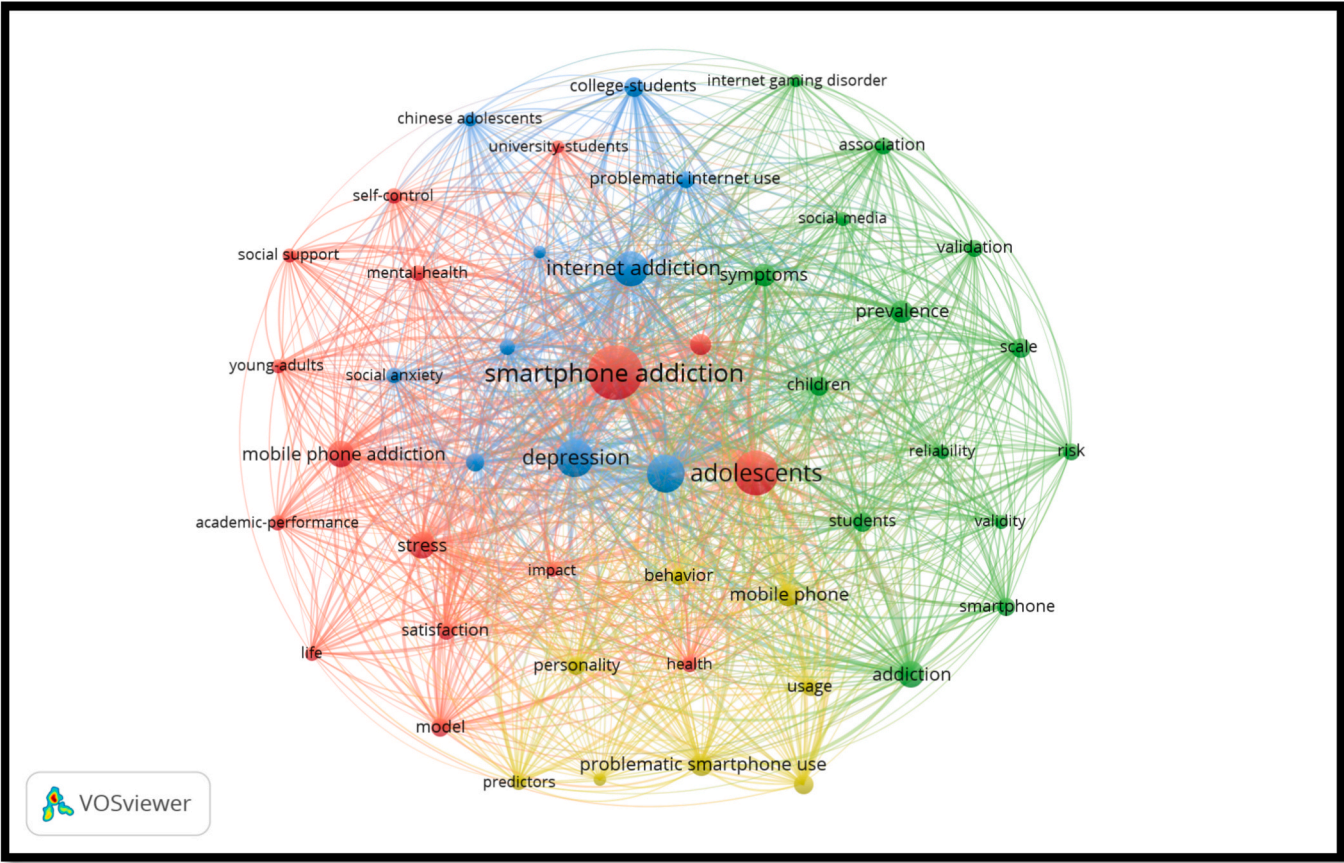


Fig. 3. Co-word analysis on smartphone addiction among youngsters.

impact on higher education student smartphone addiction. Nomophobia is conceptualized as NO MO(bile) PHO(ne) (pho) BIA, termed as anxiety or fear when one is separated from their smartphone [53]. Common nomophobia symptoms include agitation, trembling, perspiration, disorientation and respiratory alterations [54]. Nomophobia disorders are higher in teens, particularly in higher education and college students [55]. Smartphone addiction was associated with depression and was found to be higher among students with low academic success and high income levels (Sarman & Çiftci, 2024). Female students scored higher in smartphone addiction than males, suggesting a gender-based mitigation strategy.

• Cluster 4 (yellow): **Personality and communication in problematic smartphone use**

The individual personality is a matter of interest among researchers, particularly in young consumers' communication abilities and societal adaptability. Przepiorka et al. [56] found that girls use smartphone applications such as text messages, talking, social networking, organizers and cameras more often than boys. Problematic smartphone use was found to be positive toward neuroticism and negative toward conscientiousness [57]. Within the HEXACO framework, conscientiousness, openness, agreeableness and honesty-humility showed moderate negative correlations with problematic smartphone use. Smartphone addiction was also predicted to cause loneliness among undergraduate students, leading to their lack of communication [58]. A study on the big five personality traits and the dark tetrad personality could provide an overview of the essence of personality towards smartphone addiction. A summary of the co-word analysis is presented in Table 5.

4. Implications

4.1. Theoretical implications

Theoretically, this study found that most clusters suggest smartphone addiction's negative impact. This is evident in Cluster 1 (Negative consequences of smartphone addiction among youngsters), Cluster 3 (Smartphone and social media addiction) in bibliographic coupling, and Clusters 1,2 and 3 in co-word analysis. Smartphone and social media addiction have been discussed together as technology evolved [59]. Both of these addictions form a similar dynamic towards negative consequences among young consumers. The younger the user's smartphone addiction, the higher the severity of one psychological and physiological impact. Studies have explored the relationship between smartphone addiction and its consequences on different age groups. The two clusters show that the two most vulnerable groups affected by smartphone addiction are children (2 to 12 years old) and adolescents (13 to 18 years old). The impact on children includes sleeping disorders, obesity, poor vision, psychological dependence and musculoskeletal discomfort [52,51,3]. While the impact on adolescents is more common, problems exist among teenagers. Negative consequences include wellbeing, life satisfaction, and quality of life [14]. These impacts on children and adolescents should be further investigated based on different symptoms, interventions, and mitigation strategies deemed the best for different age levels.

The adoption of smartphones is due to the adoption of technology by the current generation. Young consumers, especially Generation Z and alpha, have lived with smartphones since they were toddlers. The features in smartphones make it much easier for today's generation to adopt and use smartphones, unlike the previous generation. Studies have shown that factors like compatibility with current lifestyle, perceived ease of use, and user-friendly interface designs significantly drive smartphone adoption among young consumers, making it almost second nature to Gen Z and Gen Alpha [60]. As technology evolved, advanced features like AI-powered cameras, faster chipsets, and voice

Table 5
Summary of co-word analysis on smartphone addiction among young consumers.

No.	Cluster label	Number of keywords	Representative Keywords
1 (red)	The Impact of Smartphone Addiction on Adolescents	16	Smartphone addiction, adolescents, stress, satisfaction, sleep quality, academic performance
2 (green)	Risk factors and symptoms of smartphone addiction in children	14	Children, symptoms, prevalence, risk, association, validation
3 (blue)	Smartphone Addiction and Higher Education Students	10	Depression, anxiety, loneliness, social anxiety, self-esteem, college students
4 (Yellow)	Personality and communication in problematic smartphone use	8	Problematic smartphone use, predictors, communication, personality, mobile phone, usage

assistants are becoming deciding factors when selecting a phone, making these advanced functions smoother and more intuitive [61]. These factors illustrate that technological innovations and user-oriented designs lowered barriers to adoption, embedding smartphones as an integral part of everyday life for the younger generation.

Another fundamental theoretical basis of smartphone addiction among youngsters is the dependence on parental supervision. Parents who pay attention to gadgets, particularly smartphones, would be negligent in having a face-to-face conversation with their children, characterized as phubbing [62]. Parental phubbing is the most studied phubbing type in the literature, as parents significantly influence children's growth and wellbeing. When parents focus on their digital devices over children, it will lead to children feeling neglected, lowering their social-emotional development [63]. Niu et al. [41] discovered that the parent-child relationship mediated the relationship between parental phubbing and children's smartphone addiction. Bai et al. [64] found that mother phubbing is the most critical within parental phubbing as it leads to academic burnout among children. Parental phubbing was also found to influence other negative outcomes, such as reduced social skills [63] and cyberbullying Wang et al. [65].

4.2. Managerial implications

A social marketing campaign from the government should be strategically planned and implemented to reduce smartphone dependence among youngsters. The general population has treated smartphones as part of the social landscape, taking them for granted [66]. Social marketers must understand consumer behavior, particularly their change barriers, to attain positive and effective behavioral change. Social marketing should possess the service marketing 7Ps to gauge and engage consumers' attention. For instance, the adverse impact of excessive smartphone use should be advertised, similar to certain countries' social marketing in selling cigarettes, exposing the risk of cancer [67]. It is a form of guilt appeal, a common tactic used in social marketing for smoking reduction [68]. Similar to anti-smoking campaigns, anti-smartphone addiction should employ graphic images of problematic smartphone usage as they convey effective messages rather than text information. Smartphone addiction's adverse impact could be portrayed in terms of its negative consequences, such as impacting family relationships that could lead to loss of love and distancing relationships among family members. These negative impression cues lead to consumers realizing the consequences of smartphone addiction as severe as substance addiction, such as drugs and opium.

Parents have to play their part. With knowledge, parents should know the impact of smartphone addiction on their children's psychological development. It is imperative to elicit the correct communication strategy towards parents rather than children. These emotional adverts resort towards highlighting the negative consequences caused by smartphone addiction on youngsters and providing informative knowledge to parents on how to mitigate this issue. With enough knowledge and information, to combat excessive smartphone addiction can be employed using technological intervention. Parents should employ proprietary apps such as Digital Wellbeing and Screen Time to monitor their children's daily smartphone use [69]. For older children, self-prevention apps can be utilized to manage smartphone usage through personalized interventions. For instance, Lee et al. [70] proposed a system called MindsCare that features smartphone usage analysis, monitoring functions and self-management to prevent smartphone addiction and problematic use. It was shown that the self-management system improves usability through customized intervention, which was missing in previous smartphone use management systems. This integrated and comprehensive system provides an automatic and manual intervention system and a platform to communicate with clinicians. With advanced digital technology and innovative intervention strategies, smartphone addiction can be minimized through certain control measures that are practical within the household context. On the

other hand, the government could develop a policy or guidelines parents should abide by to ensure smartphone use at home is used only for necessity.

5. Limitations and future research avenues

This review possesses several limitations. First, the database only includes within 5 years, between 2019 and 2024. Limiting the years would risk missing some clusters prevalent not within the 5 years. However, limiting it to only 5 years enables focus on more recent and contemporary themes on the topic. Secondly, studies on smartphone addiction are internationally recognized in languages other than English in various parts of the world. As most studies are indexed in reputable indexing bodies such as WoS and Scopus, relevant papers might be missing in other languages. Hence, future studies could adopt different methods, such as a systematic or scoping review in another database to capture publications written in non-English languages. Third, this study focuses on the specific wording of "smartphone addiction" and "mobile addiction" without looking into other relevant keywords in the literature, such as "problematic smartphone use" or "maladaptive smartphone use". Furthermore, this study focuses on young consumers and only includes three segments of youngsters: children, adolescents and teenagers. Another age group that constitutes young is "young adults". Young adults can be described as aged between 18 and 25, sometimes extending to 29 [71]. However, we excluded young adults to ensure the exclusivity of this study's focus on young consumers. Future studies could explore the thematic structure of young adults' smartphone addiction.

6. Conclusion

This study has provided an indispensable overview of smartphone addiction among young consumers in today's digitalized environment. Several themes are produced by employing two science mapping analyses through a bibliometric approach. In bibliographic coupling, four themes were produced: 1) Negative consequences of smartphone addiction among youngsters, 2) Parental phubbing and impact on adolescent smartphone addiction, 3) Smartphone and social media addiction, and 4) Smartphone addiction scale. In comparison, the themes in co-word analysis are: 1) The Impact of Smartphone Addiction on Adolescents, 2) Risk factors and symptoms of smartphone addiction in children, 3) Smartphone Addiction and higher education students: Depression, Anxiety, and Self-Esteem, and 4) Personality and communication in problematic smartphone use. Future trends are associated with the risk and impact of smartphone addiction among young consumers. As serious as this problem is at the family and societal level, it should be managed at the national level, as societal issues would eventually impact national progress and development.

CRediT authorship contribution statement

Muhammad Ashraf Fauzi: Writing – review & editing, Writing – original draft, Visualization, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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